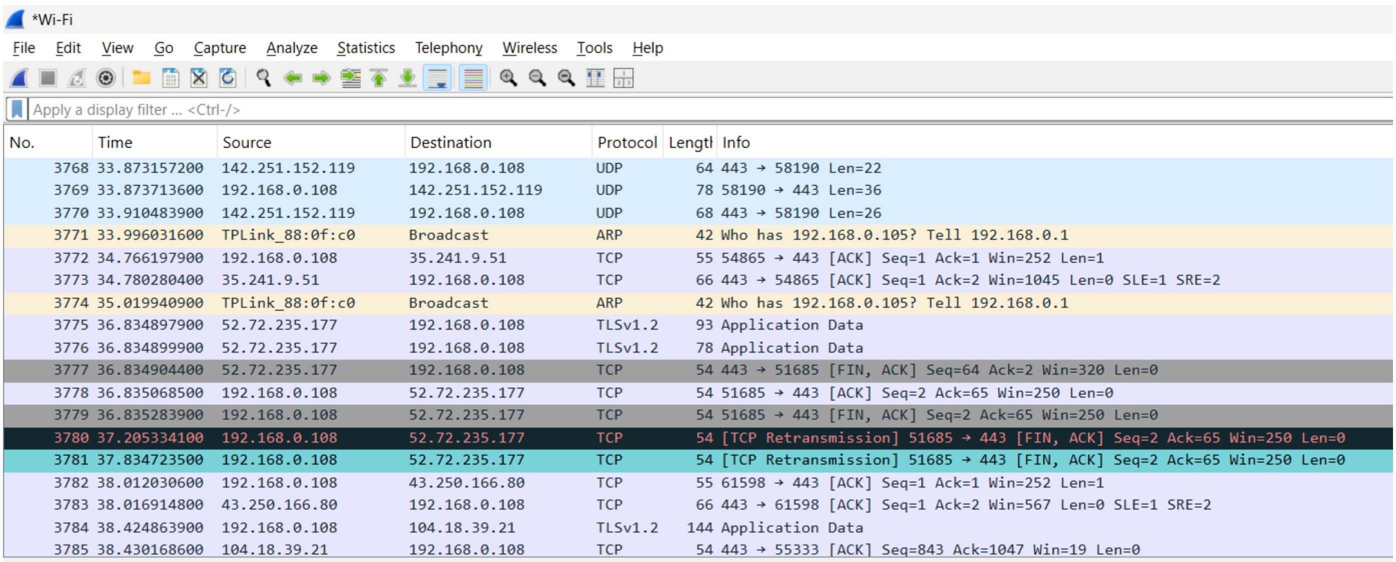
Day 1

Introduction to Networking and the OSI Model Theory (1 hour):What is Networking? Overview of the OSI Model: Layers and their functions. Practical (1 hour):Use a network sniffer (e.g., Wireshark) to observe traffic and identify layers. Task: Create a diagram of the OSI Model with examples for each layer.

1. **Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.**

ANS :

No.	Protocol	Purpose
1	DNS	Converts website names into IP addresses
2	TCP	Reliable data transport
3	TLSv1.3	Secure encrypted communication
4	HTTP/HTTPS	Website communication
5	ARP	Finds MAC addresses on local network
6	ICMP	Network diagnostics
7	UDP	Fast communication protocol



No.	Time	Source	Destination	Protocol	Lengtl	Info
3768	33.873157200	142.251.152.119	192.168.0.108	UDP	64	443 → 58190 Len=22
3769	33.873713600	192.168.0.108	142.251.152.119	UDP	78	58190 → 443 Len=36
3770	33.910483900	142.251.152.119	192.168.0.108	UDP	68	443 → 58190 Len=26
3771	33.996031600	TPLink_88:0f:c0	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
3772	34.766197900	192.168.0.108	35.241.9.51	TCP	55	54865 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
3773	34.780280400	35.241.9.51	192.168.0.108	TCP	66	443 → 54865 [ACK] Seq=1 Ack=2 Win=1045 Len=0 SLE=1 SRE=2
3774	35.019940900	TPLink_88:0f:c0	Broadcast	ARP	42	Who has 192.168.0.105? Tell 192.168.0.1
3775	36.834897900	52.72.235.177	192.168.0.108	TLSv1.2	93	Application Data
3776	36.834899900	52.72.235.177	192.168.0.108	TLSv1.2	78	Application Data
3777	36.834904400	52.72.235.177	192.168.0.108	TCP	54	443 → 51685 [FIN, ACK] Seq=64 Ack=2 Win=320 Len=0
3778	36.835068500	192.168.0.108	52.72.235.177	TCP	54	51685 → 443 [ACK] Seq=2 Ack=65 Win=250 Len=0
3779	36.835283900	192.168.0.108	52.72.235.177	TCP	54	51685 → 443 [FIN, ACK] Seq=2 Ack=65 Win=250 Len=0
3780	37.205334100	192.168.0.108	52.72.235.177	TCP	54	[TCP Retransmission] 51685 → 443 [FIN, ACK] Seq=2 Ack=65 Win=250 Len=0
3781	37.834723500	192.168.0.108	52.72.235.177	TCP	54	[TCP Retransmission] 51685 → 443 [FIN, ACK] Seq=2 Ack=65 Win=250 Len=0
3782	38.012030600	192.168.0.108	43.250.166.80	TCP	55	61598 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
3783	38.016914800	43.250.166.80	192.168.0.108	TCP	66	443 → 61598 [ACK] Seq=1 Ack=2 Win=567 Len=0 SLE=1 SRE=2
3784	38.424863900	192.168.0.108	104.18.39.21	TLSv1.2	144	Application Data
3785	38.430168600	104.18.39.21	192.168.0.108	TCP	54	443 → 55333 [ACK] Seq=843 Ack=1047 Win=19 Len=0

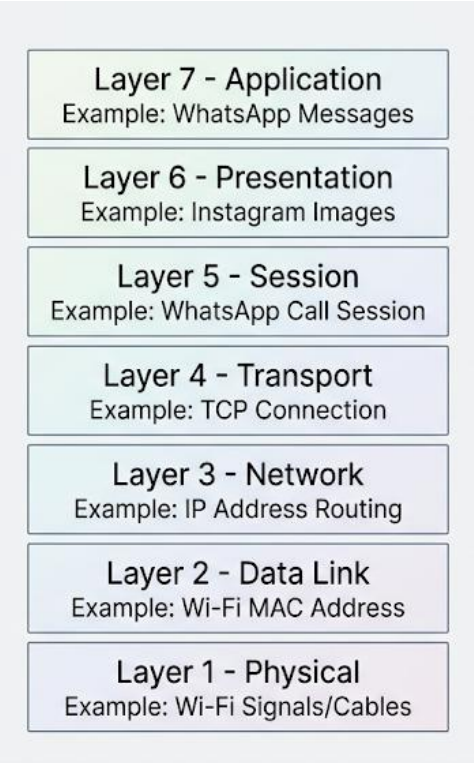
2. **Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Hint : Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.**

ANS :

Packet	Protocol	Main OSI Layer	Explanation
Packet 1	DNS	Application Layer (Layer 7)	Resolves instagram.com into an IP address
Packet 2	TCP	Transport Layer (Layer 4)	Establishes reliable connection between devices
Packet 3	ARP	Data Link Layer (Layer 2)	Finds the MAC address of a device on the network

3. **Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).**

ANS :



4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Hint : Think about how your message is typed, packaged, sent over the internet, and received.

ANS :

Explanation

When I send a message on WhatsApp, it travels through several layers of the OSI Model.

1. Application Layer (Layer 7)

I type a message such as "Hello". WhatsApp creates the message and prepares it to be sent.

2. Presentation Layer (Layer 6)

The message is encrypted so that only the intended recipient can read it. This provides security and privacy.

3. Session Layer (Layer 5)

WhatsApp maintains a communication session between my phone and the recipient's device, ensuring both devices stay connected.

4. Transport Layer (Layer 4)

TCP divides the message into smaller segments and ensures they are delivered reliably and in the correct order.

5. Network Layer (Layer 3)

IP addresses are added, and routers determine the best path for the message across the Internet.

6. Data Link Layer (Layer 2)

The message is packaged into frames with MAC addresses so it can travel within the local network.

7. Physical Layer (Layer 1)

The data is converted into electrical signals or Wi-Fi radio waves and transmitted through the network.

Conclusion

The recipient's phone receives the signals and the process happens in reverse until the WhatsApp message appears on their screen.